

MATH-1014 T10B

Midterm Exercise (2)

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Polar Curves: Arc Length

Problem 11

A curve in the xy -plane is defined by the polar equation $r = \sqrt{\sec 2\theta}$, where $0 \leq \theta \leq \frac{\pi}{6}$. which of the following definite integrals is the arc length of the curve?

Hint for Solution:

Recall the arc length formula for a curve given in polar coordinates $r = f(\theta)$ from $\theta = a$ to $\theta = b$:

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

You will need to compute $\frac{dr}{d\theta}$ using the chain rule and substitute it into the formula.

Solution to Problem 11

Arc Length Formula: $L = \int_a^b \sqrt{r^2 + (r')^2} d\theta$

1. **Derivative (r'):**

$$r = (\sec 2\theta)^{\frac{1}{2}} \implies r' = \frac{1}{2}(\sec 2\theta)^{-\frac{1}{2}}(2 \sec 2\theta \tan 2\theta) = \sqrt{\sec 2\theta} \tan 2\theta$$

2. **Integrand Simplification:**

$$\begin{aligned} r^2 + (r')^2 &= \sec 2\theta + \sec 2\theta \tan^2 2\theta \\ &= \sec 2\theta(1 + \tan^2 2\theta) = \sec^3 2\theta \end{aligned}$$

3. **Result:**

$$L = \int_0^{\frac{\pi}{6}} \sqrt{\sec^3 2\theta} d\theta = \int_0^{\frac{\pi}{6}} (\sec 2\theta)^{\frac{3}{2}} d\theta$$

Polar Curves: Area

Problem 14 (a) & (b)

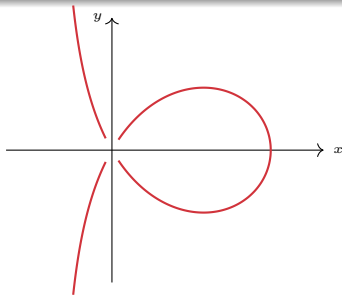
[15 pts] Consider the curve defined by the equation $y^2 = x^2 \left(\frac{3-x}{1+x} \right)$.

(a) Show that the polar equation of the curve can be written as

$$r = (3 \cos^2 \theta - \sin^2 \theta) \sec \theta. \quad [4 \text{ pts}]$$

(b) Find the area inside the loop of the given curve. [7 pts]

(Hint: At what θ will the curve hit the origin?)

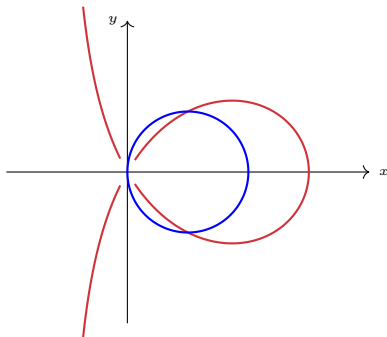


Polar Curves: Area between curves

Problem 14 (c)

(c) Use integral(s) to express the area inside the loop of the given curve but outside the circle given by the polar equation $r = 2 \cos \theta$. **Do not evaluate the integral.** [4 pts]

(Hint: At what θ will the curves intersect each other?)



Solution to 14 (a) & (b)

(a) Polar Form: Let $x = r \cos \theta$, $y = r \sin \theta$.

$$r^2 \sin^2 \theta (1 + r \cos \theta) = r^2 \cos^2 \theta (3 - r \cos \theta)$$

$$r \cos \theta (\sin^2 \theta + \cos^2 \theta) = 3 \cos^2 \theta - \sin^2 \theta$$

$$r \cos \theta = 3 \cos^2 \theta - \sin^2 \theta \implies r = (3 \cos^2 \theta - \sin^2 \theta) \sec \theta$$

Alternative simplified form: $r = 4 \cos \theta - \sec \theta$.

(b) Loop Area: Hits origin when $r = 0 \implies 4 \cos^2 \theta = 1 \implies \theta = \pm \frac{\pi}{3}$.

$$\begin{aligned} \text{Area} &= 2 \times \frac{1}{2} \int_0^{\frac{\pi}{3}} (4 \cos \theta - \sec \theta)^2 d\theta \\ &= \int_0^{\frac{\pi}{3}} (8 \cos 2\theta + \sec^2 \theta) d\theta \quad (\text{after simplifying}) \\ &= [4 \sin 2\theta + \tan \theta]_0^{\frac{\pi}{3}} = \boxed{3\sqrt{3}} \end{aligned}$$

Solution to 14 (c)

1. Intersection Points:

Set $r_{\text{curve}} = r_{\text{circle}}$:

$$4 \cos \theta - \sec \theta = 2 \cos \theta$$

$$2 \cos \theta = \sec \theta \implies 2 \cos^2 \theta = 1 \implies \theta = \pm \frac{\pi}{4}$$

2. Area Integral:

Using symmetry across the polar axis:

$$\text{Area} = 2 \times \frac{1}{2} \int_0^{\frac{\pi}{4}} (r_{\text{curve}}^2 - r_{\text{circle}}^2) d\theta$$

$$\text{Area} = \boxed{\int_0^{\frac{\pi}{4}} \left((4 \cos \theta - \sec \theta)^2 - (2 \cos \theta)^2 \right) d\theta}$$

Reduction Formulas (1)

Part III

Problem 1 (a) & (b)

([25 points]) Let $I_n = \int_0^2 \frac{1}{(x^2 + 4)^n} dx$, where $n = 1, 2, 3, \dots$ is a positive integer.

(a) Using integration by parts, or otherwise, find $A(n), B(n)$, which are expressions depending on n , such that

$$I_{n+1} = A(n)I_n + B(n).$$

(Hint: Start with I_n .)

[12 pts]

(b) Using (a), or otherwise, evaluate the integral

[7 pts]

Numerical Approximation

Problem 1 (c)

(c) If Simpson's rule on four subintervals is used to approximate

$$\pi = \int_0^2 \frac{8}{x^2 + 4} dx,$$

a rational approximate value of π can be found as

$$\pi \approx \frac{1}{3} \left[1 + \frac{64}{a} + \frac{8}{b} + \frac{64}{c} + \frac{1}{d} \right]$$

where a, b, c, d are positive integers. Find a, b, c, d .

[6 pts]

Reduction Formulas (2)

Problem 13 (a) & (b)

[20 pts] Let $I_n = \int_3^8 \frac{1}{x^n \sqrt{x+1}} dx$, where $n \geq 0$ is an integer.

(a) Evaluate I_0 . [3 pts]

(b) Evaluate I_1 . [6 pts]

Reduction Formulas (2)

Problem 13 (c) & (d)

(c) Apply integration by parts to show that

[8 pts]

$$I_{n+1} = \frac{1-2n}{2n}I_n - \frac{3}{n8^n} + \frac{2}{n3^n}.$$

(d) Evaluate I_2 .

[3 pts]

Comprehensive Problem (Part 1)

Part III

(a) & (b)

([25 points]) Let $I_n = \int_0^{\frac{\pi}{3}} \tan^n \theta \sec^3 \theta d\theta$, where $n = 0, 1, 2, \dots$ is an integer.

(a) By writing $\tan^n \theta \sec^3 \theta = \tan^{n-1} \theta \cdot (\tan \theta \sec^3 \theta)$ and using integration by parts, show that for $n \geq 2$,

$$I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$$

[10 pts]

(b) Evaluate I_0 .

[4 pts]

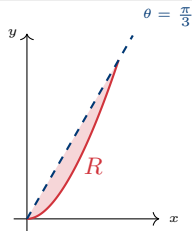
(Hint: You may need the integral of $\sec \theta$.)

Comprehensive Problem (Part 2)

(c) & (d)

(c) The region R in the first quadrant is enclosed by the polar curve $r = \sqrt{2 \tan^2 \theta \sec^3 \theta}$, the ray $\theta = \frac{\pi}{3}$, and the polar axis ($\theta = 0$). Evaluate the exact area of R . [6 pts]

(d) If Simpson's rule on two subintervals is used to approximate I_0 , the approximate value can be written as $I_0 \approx \frac{\pi}{a} \left[b + \frac{c\sqrt{3}}{9} \right]$. Find $a, b, c \in \mathbb{Z}^+$. [5 pts]



Solution to (a) - Part 1

Show: $I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$

Let $u = \tan^{n-1} \theta \implies du = (n-1) \tan^{n-2} \theta \sec^2 \theta d\theta$.

Let $dv = \tan \theta \sec^3 \theta d\theta \implies v = \frac{1}{3} \sec^3 \theta$.

Apply Integration by Parts ($\int u dv = uv - \int v du$):

$$I_n = \left[\frac{1}{3} \tan^{n-1} \theta \sec^3 \theta \right]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \frac{1}{3} \sec^3 \theta \cdot (n-1) \tan^{n-2} \theta \sec^2 \theta d\theta$$

1. Evaluate the boundary term: At $\theta = \frac{\pi}{3}$, $\tan(\frac{\pi}{3}) = \sqrt{3}$ and $\sec(\frac{\pi}{3}) = 2$.
At $\theta = 0$, $\tan(0) = 0$ (since $n \geq 2$).

$$\text{Boundary} = \frac{1}{3} (\sqrt{3})^{n-1} (2)^3 - 0 = \frac{8}{3} (\sqrt{3})^{n-1}$$

Solution to (a) - Part 2

2. Simplify the integral term: Replace $\sec^2 \theta$ with $1 + \tan^2 \theta$:

$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3} \int_0^{\frac{\pi}{3}} \tan^{n-2} \theta \sec^3 \theta (1 + \tan^2 \theta) d\theta$$

$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3} (I_{n-2} + I_n)$$

Multiply the entire equation by 3 to clear denominators:

$$3I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} - (n-1)I_n$$

$$(3+n-1)I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2}$$

$$(n+2)I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2}$$

Divide by $(n+2)$ to obtain the final formula:

$$I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$$

Solution to (b) & (c)

(b) Evaluate I_0 :

$$\begin{aligned} I_0 &= \int_0^{\frac{\pi}{3}} \sec^3 \theta \, d\theta = \frac{1}{2} \left[\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta| \right]_0^{\frac{\pi}{3}} \\ &= \frac{1}{2} \left(2\sqrt{3} + \ln(2 + \sqrt{3}) \right) - 0 = \boxed{\sqrt{3} + \frac{1}{2} \ln(2 + \sqrt{3})} \end{aligned}$$

(c) Evaluate Polar Area:

$$A = \frac{1}{2} \int_0^{\frac{\pi}{3}} r^2 \, d\theta = \frac{1}{2} \int_0^{\frac{\pi}{3}} 2 \tan^2 \theta \sec^3 \theta \, d\theta = \int_0^{\frac{\pi}{3}} \tan^2 \theta \sec^3 \theta \, d\theta$$

This is exactly I_2 . Apply the reduction formula from (a) with $n = 2$:

$$\begin{aligned} I_2 &= \frac{8(\sqrt{3})^1}{4} - \frac{1}{4} I_0 = 2\sqrt{3} - \frac{1}{4} \left(\sqrt{3} + \frac{1}{2} \ln(2 + \sqrt{3}) \right) \\ &= \boxed{\frac{7\sqrt{3}}{4} - \frac{1}{8} \ln(2 + \sqrt{3})} \end{aligned}$$

Solution to (d)

(d) Simpson's Rule for I_0 : Approximate $\int_0^{\frac{\pi}{3}} \sec^3 \theta d\theta$ with 2 subintervals ($n = 2$).

Step size: $\Delta x = \frac{\pi/3 - 0}{2} = \frac{\pi}{6}$.

Points: $x_0 = 0$, $x_1 = \frac{\pi}{6}$, $x_2 = \frac{\pi}{3}$.

Function values $f(\theta) = \sec^3 \theta$:

- $f(0) = 1^3 = 1$
- $f(\frac{\pi}{6}) = (\frac{2}{\sqrt{3}})^3 = \frac{8}{3\sqrt{3}} = \frac{8\sqrt{3}}{9}$
- $f(\frac{\pi}{3}) = 2^3 = 8$

Simpson's Formula: $S_2 = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + f(x_2)]$

$$S_2 = \frac{\pi/6}{3} \left[1 + 4 \left(\frac{8\sqrt{3}}{9} \right) + 8 \right] = \frac{\pi}{18} \left[9 + \frac{32\sqrt{3}}{9} \right]$$

Matching this with $\frac{\pi}{a} \left[b + \frac{c\sqrt{3}}{9} \right]$, we get:

$$a = 18, \quad b = 9, \quad c = 32$$